

MACHINE LEARNING & STATISTICAL LEARNING — MODULE: STATISTICAL LEARNING

Statistical Methods

A Supervised and Unsupervised Statistical Learning Analysis in R

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Research questions



Supervised — Classification

RQ1

Which clinical variables appear to be associated with thyroid cancer recurrence?

H1

Recurrence is associated with clinical response and other clinical characteristics.

Dataset: Differentiated Thyroid Cancer Recurrence (UCI ML Repository)



Unsupervised — Clustering

RQ2

Can observations in the UR3 CobotOps dataset be grouped into meaningful clusters?

H2

The dataset contains underlying patterns identifiable using clustering methods.

Dataset: UR3 CobotOps — sensor & operational time series (UCI ML Repository)

Datasets & Preprocessing



Supervised — Already Cross-Sectional

384

patients in the dataset, target variable
recurred

- ✓ No missing values across all clinical variables
- ✓ Target variable split easily chosen
- ✓ Numerical and categorical data
- ✓ No extensive pre-processing required



Unsupervised — Transformed From Time Series

- ✓ Grouped by cycle: multiple sensor rows → one row per robot cycle
- ✓ Mean, SD, min, and max calculated per sensor variable
- ✓ Variables scaled with R's `scale()` before clustering
- ✓ `robot_protective_stop` and `grip_lost` withheld for post-hoc use only

Exploratory Analysis — Supervised

- ✓ No missing values across all 383 patient records
- ✓ Majority of patients are female
- ✓ Target is imbalanced: 275 of 383 patients had no recurrence
- ✓ Most patients have papillary thyroid cancer
- ✓ Majority of patients are classified as low risk

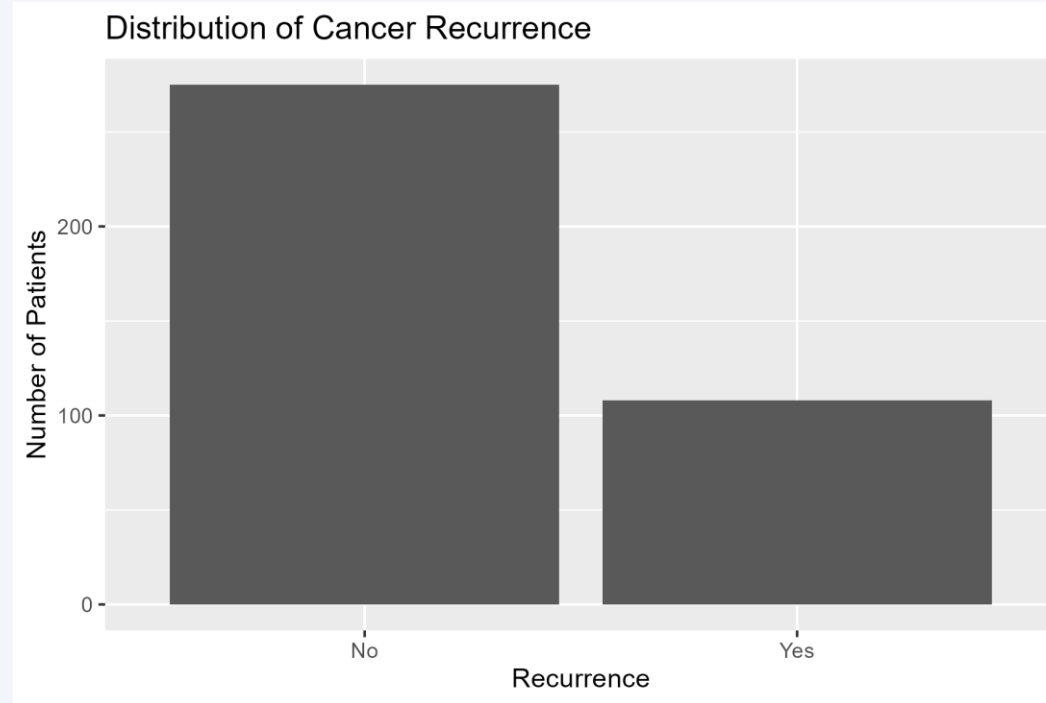


Figure 1 · Distribution of the recurrence target variable

Response and Age Track With Recurrence

These plots do not prove causality, but indicate that response and age may carry useful predictive information.

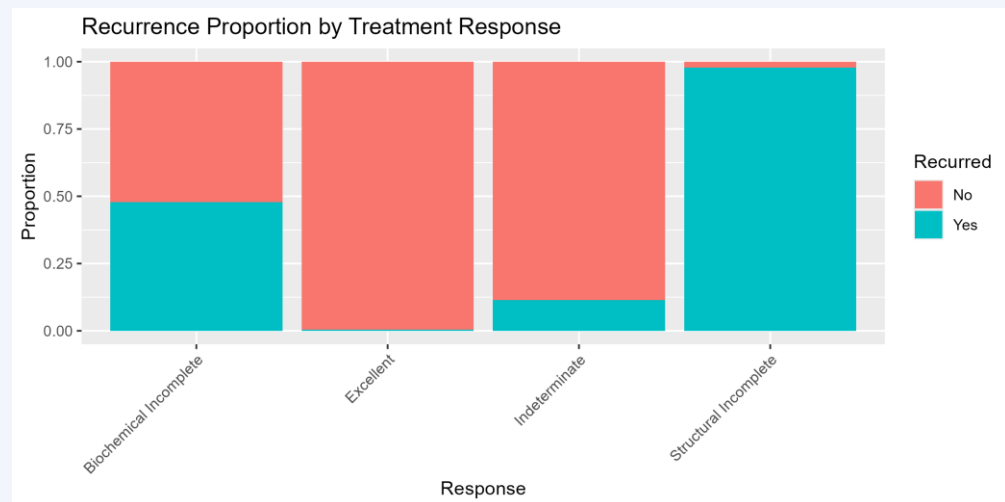


Figure 2 · Recurrence proportion by treatment response

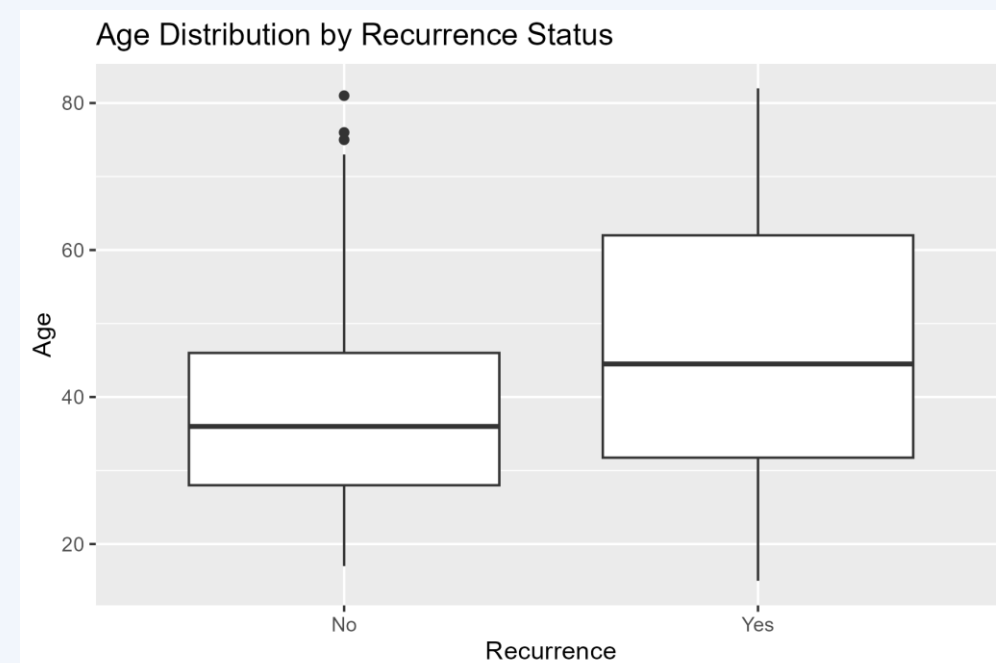


Figure 3 · Age distribution by recurrence status

Statistical Association Tests

Variable	Test used	p-value
Age	Spearman correlation ($\rho = 0.212$)	< 0.001
Response	Fisher's exact test	< 0.001
Risk	Fisher's exact test	< 0.001
Stage	Fisher's exact test	< 0.001
Adenopathy	Fisher's exact test	< 0.001
Gender	Fisher's exact test	< 0.001
Physical examination	Fisher's exact test	0.007
Thyroid function	Fisher's exact test	0.328

Only one non-significant variable

Every tested variable was significantly associated with recurrence except Thyroid function ($p = 0.328$).

8 further variables — Smoking, History of smoking/radiotherapy, Pathology, Focality, T/N/M classification — were also significant ($p < 0.05$) and were not added here for space.

Logistic Regression — Final Model

Predictors were reduced in two passes — first removing variables with extreme standard errors, then removing risk — to reach a stable, interpretable model.

Term	Estimate	Std. error	p-value
(Intercept)	-1.131	1.606	0.481
Age	0.037	0.021	0.086
Gender: M	0.416	0.766	0.587
Response: Excellent	-4.680	1.126	< 0.001
Response: Indeterminate	-2.525	0.745	< 0.001
Response: Structural Incomplete	2.947	0.927	0.001
Adenopathy: Extensive	14.079	1565.142	0.993
Adenopathy: Left	1.320	1.478	0.372
Adenopathy: No	-0.972	1.032	0.346
Adenopathy: Posterior	14.061	2779.686	0.996
Adenopathy: Right	0.515	1.208	0.670

Treatment response is the clearest signal

All three response categories remain significant after model reduction, while adenopathy and gender lose significance and show unstable, oversized standard errors.

Decision Tree — Splitting on Response

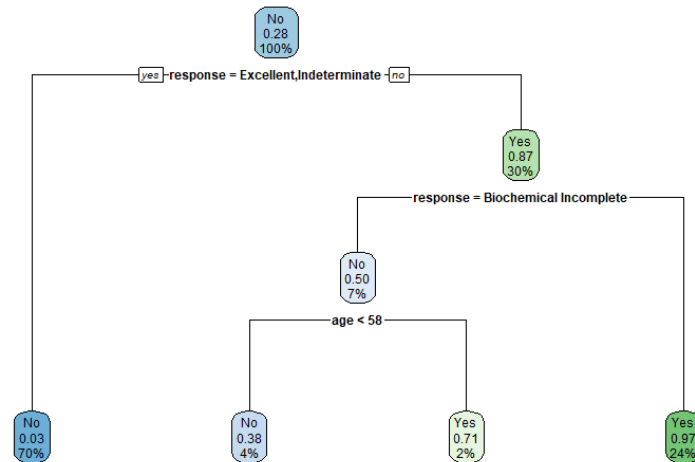
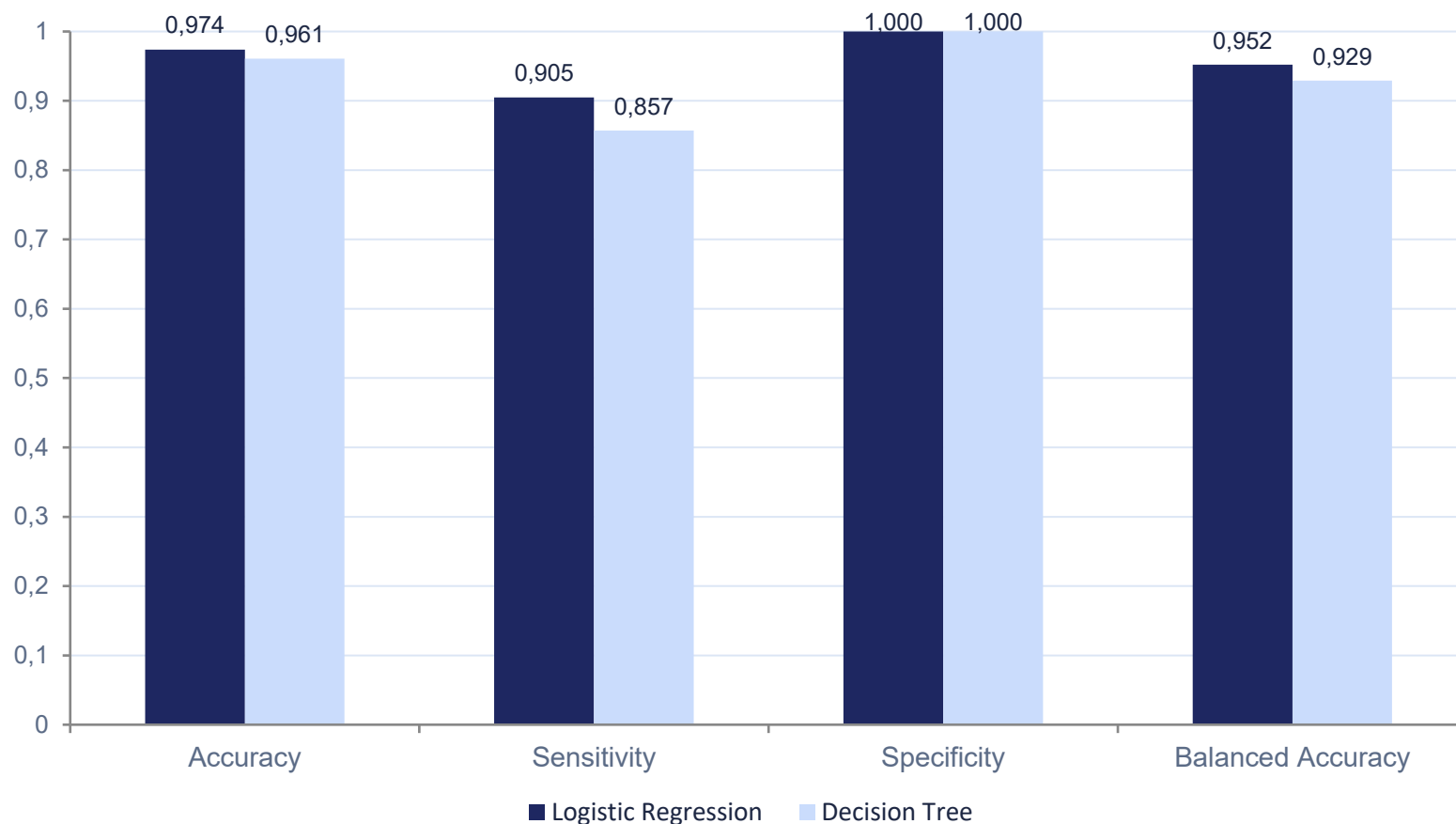


Figure 4 · Decision tree for recurrence prediction

- Treatment response is the most important splitting variable
- Excellent / Indeterminate response → mostly classified as no recurrence
- Within Biochemical Incomplete, age < 58 further separates outcomes
- Structural Incomplete response → classified almost entirely as recurrence

SECTION 6.4

Model Comparison on the Test Set



0.952

Best balanced accuracy, achieved by the reduced Logistic Regression model

Both models reach perfect specificity (1.000), but Logistic Regression identifies more true recurrence cases (sensitivity 0.905 vs. 0.857).

Supervised Analysis — Results

H1 supported

Recurrence is associated with treatment response and other clinical characteristics.

- 1 Response** Strongest and most stable predictor across all models
- 2 Risk** Significant association, but unstable in the reduced regression
- 3 Stage** Significant association; interpret with caution
- 4 Adenopathy** Significant overall, but lost stability after model reduction

Takeaway

The strongest evidence is found for treatment response. Other clinical variables show real but less stable associations once the model is reduced for interpretability.

From Time Series to Clusterable Data



robot_protective_stop and grip_lost are removed before clustering and reintroduced afterward, purely for post-hoc interpretation.

UMAP Reveals Visible Structure

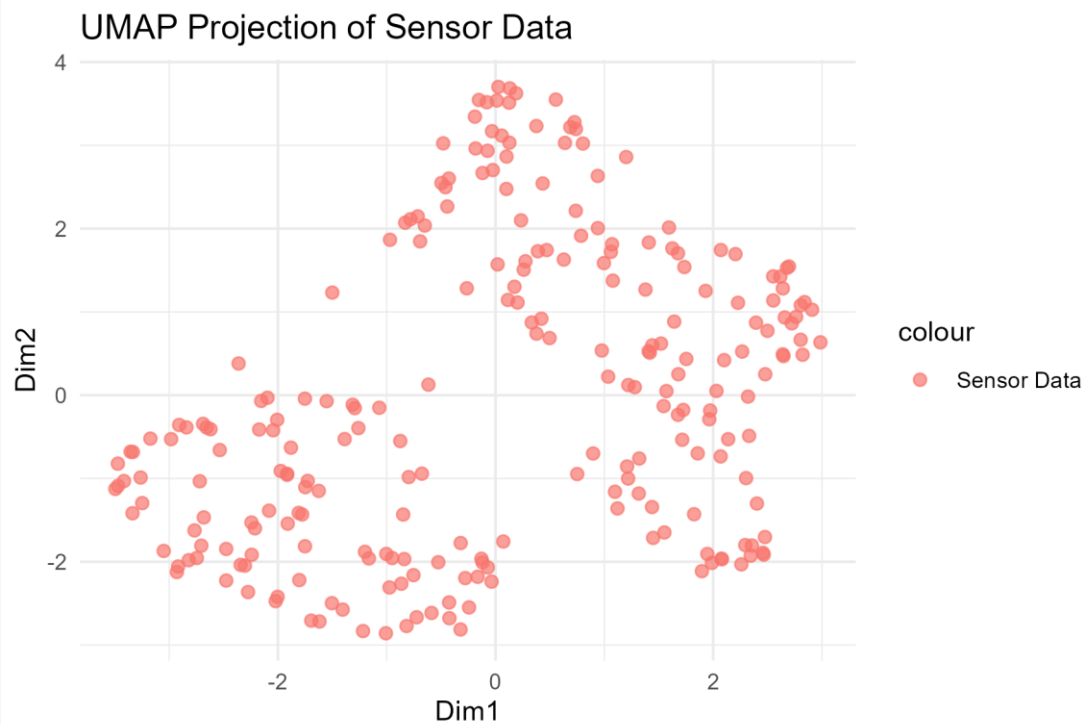


Figure 5 · UMAP projection of aggregated sensor data

Robot cycles are not randomly spread

Projected into two dimensions, the aggregated sensor variables form two visually distinct regions, suggesting genuine underlying structure rather than noise.

UMAP is used here purely as a visualization tool — clustering itself is performed in the next step with a Gaussian Mixture Model.

Coloring by Operational Events

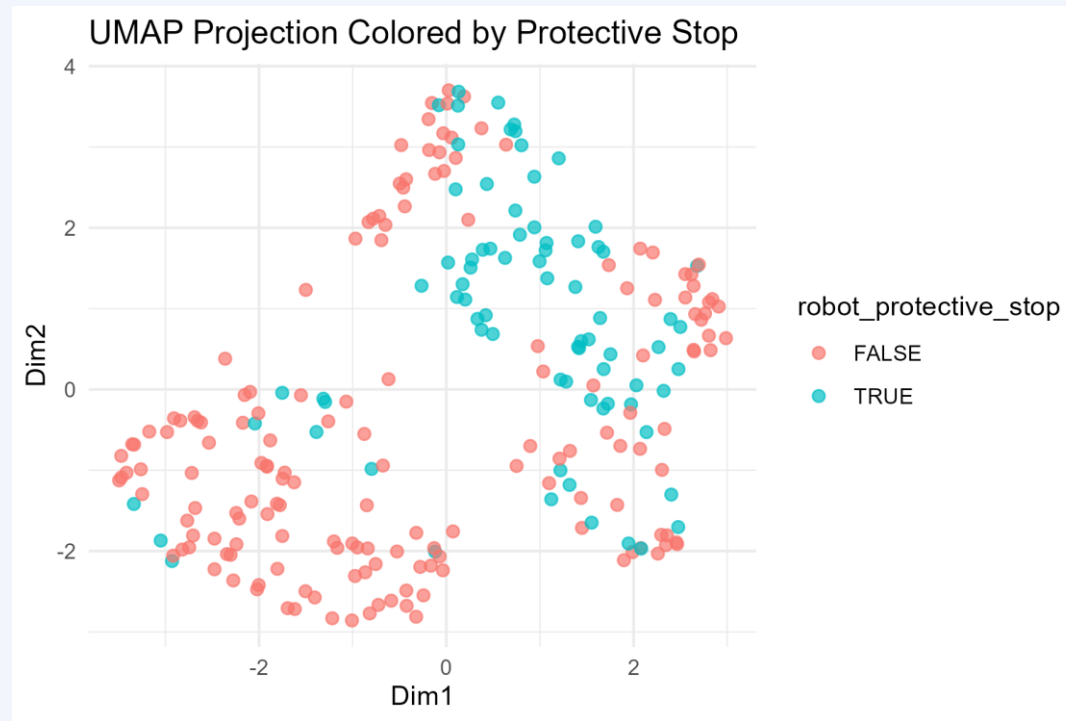


Figure 6 · Protective stop — clear grouping on the right side

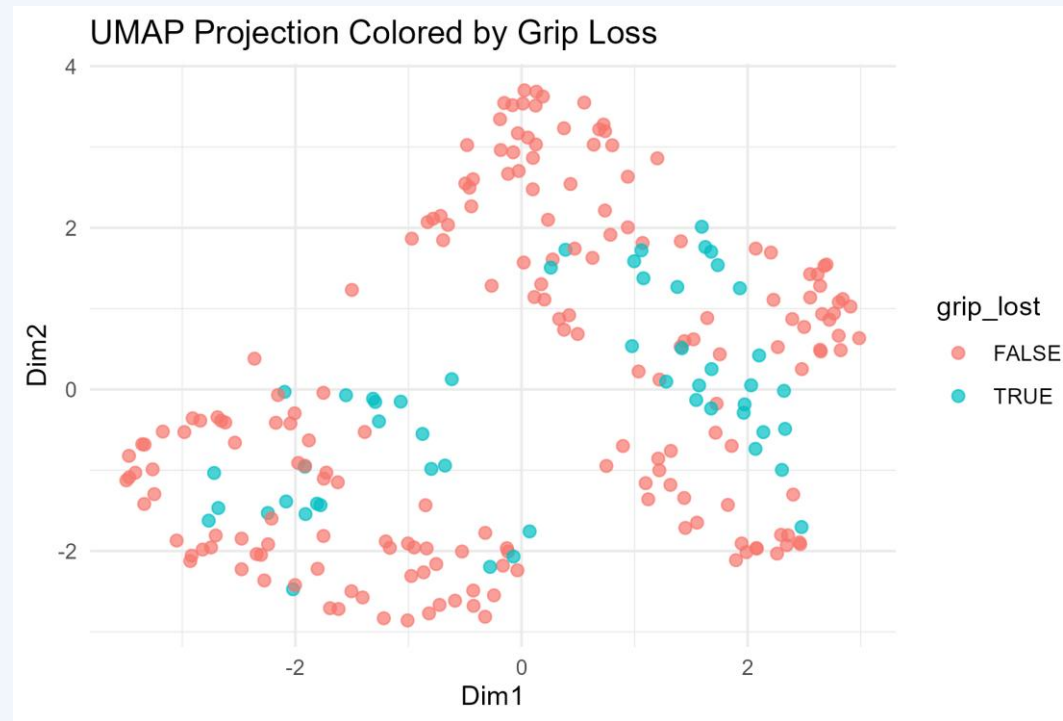


Figure 7 · Grip loss — spread loosely across several regions

Gaussian Mixture Model — Six Clusters

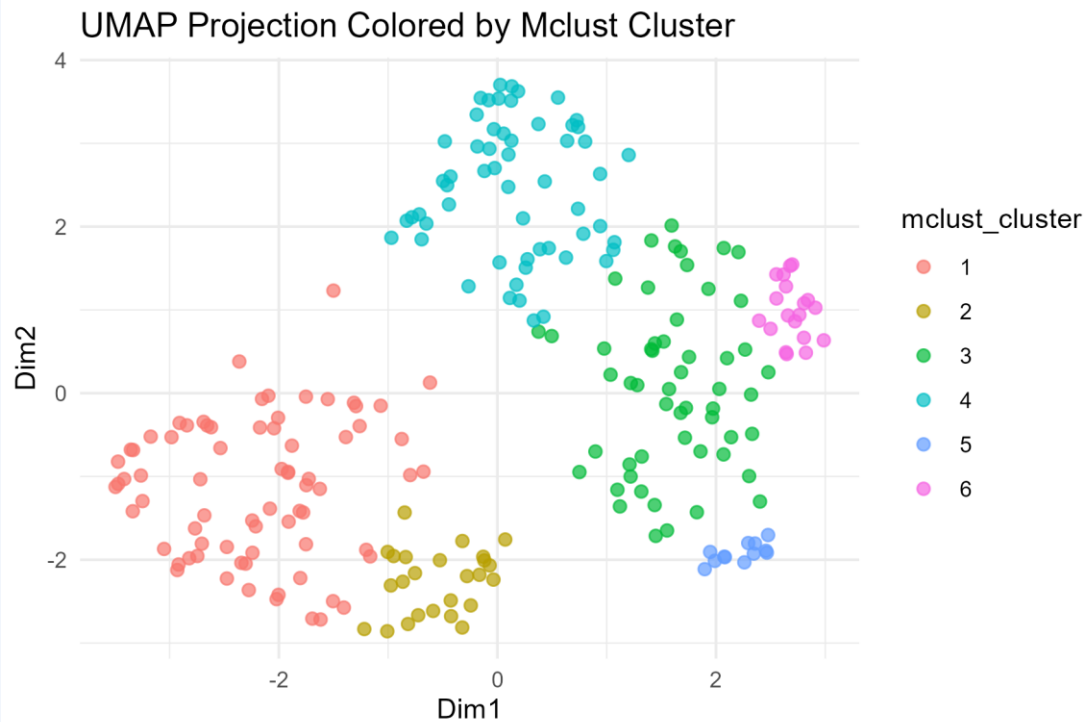


Figure 8 · UMAP projection colored by Mclust cluster

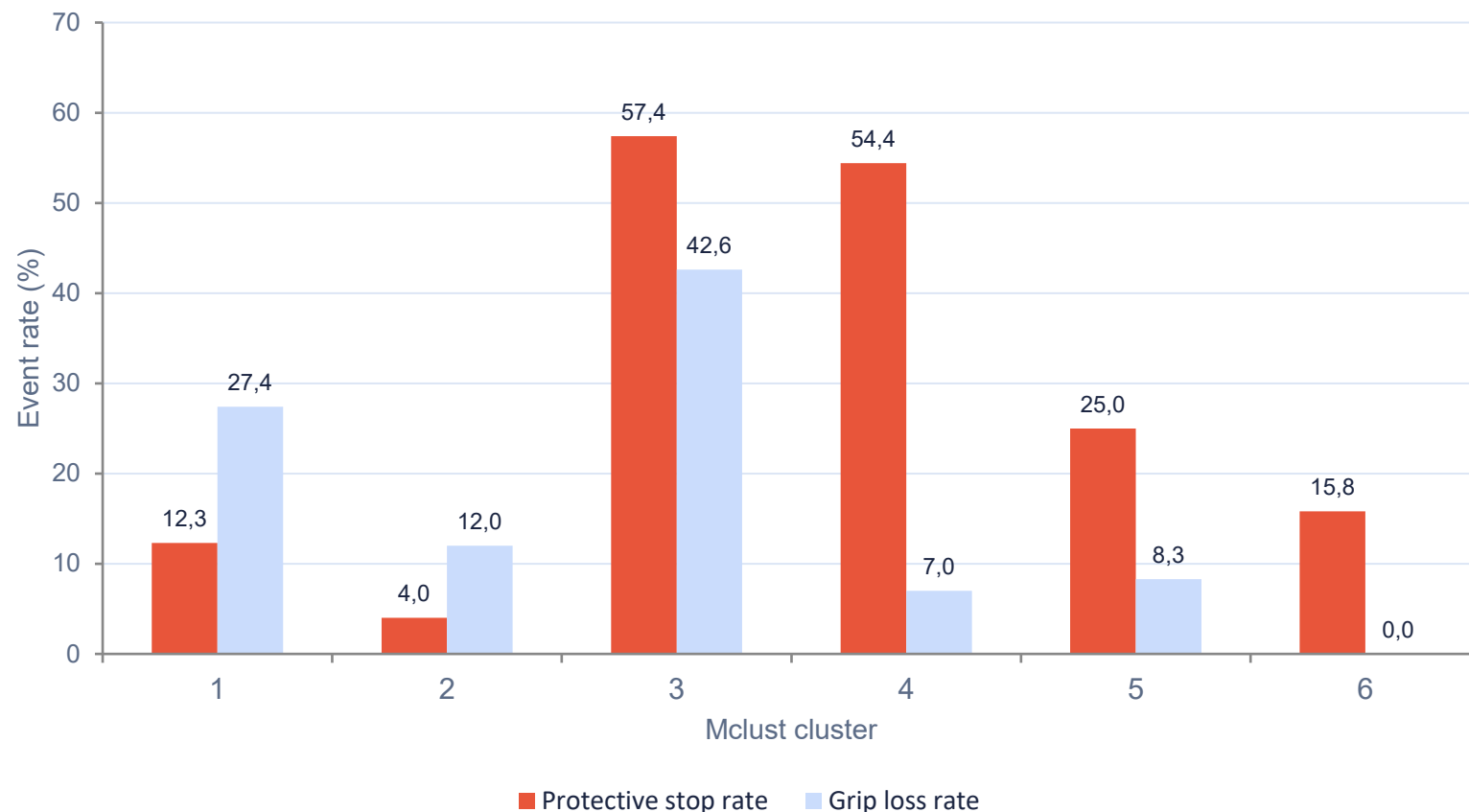


mclust::Mclust

Gaussian Mixture Models were fitted to the scaled sensor features and clearly divided the UMAP projection into six distinct clusters of robot cycles.

Each cluster groups cycles with similar sensor behavior — the next step checks whether these groupings relate to real operational events.

Which Clusters Are Most Problematic?



Cluster 3

is the most problematic group — highest protective-stop rate (57.4%) and highest grip-loss rate (42.6%).

Cluster 4 shows a high protective-stop rate (54.4%) but a low grip-loss rate (7.0%) — the two events relate to different sensor patterns.

Unsupervised Analysis — Results

H2 supported

The UR3 CobotOps dataset contains underlying patterns identifiable through clustering.

- ✓ GMM clustering divided the UMAP projection into six coherent groups of robot cycles
- ✓ Clusters were not random: event rates varied sharply across them
- ✓ Cluster 3 combines both the highest protective-stop and grip-loss rates
- ✓ Protective stops and grip losses relate to different, partly distinct sensor patterns

Takeaway

Even without using event labels for clustering, the resulting groups carry real operational meaning — strong post-hoc evidence that the clusters are meaningful, not arbitrary.

CONCLUSION

A Complete Statistical Learning Workflow

Both analyses successfully answered their research questions and supported their hypotheses.



Supervised

Treatment response, risk, stage, and adenopathy are strongly associated with thyroid cancer recurrence. A reduced logistic regression model reached 0.952 balanced accuracy, with response as the most stable predictor.

RQ1 answered · H1 supported



Unsupervised

Aggregated UR3 CobotOps sensor cycles form visible structure in UMAP space. Gaussian Mixture Model clustering produced six groups whose protective-stop and grip-loss rates differ sharply — confirming the clusters reflect real robot behavior.

RQ2 answered · H2 supported